

Spectral bisection

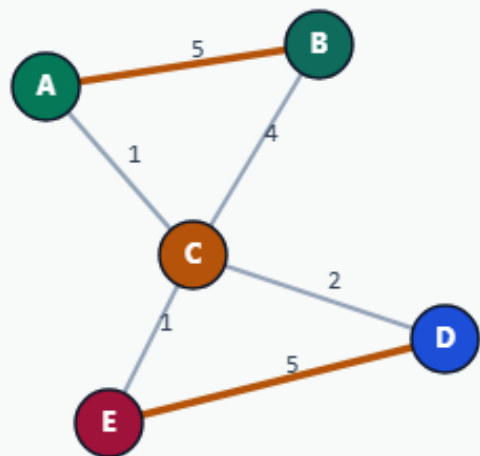
Spectral bisection uses the graph Laplacian and its Fiedler vector to propose a balanced cut

Build the Laplacian

SPECTRAL BISECTION

Build the Laplacian

Step 1 / 5 · laplacian · module02-03-spectral-bisection



Explanation

Spectral methods read connectivity from the graph Laplacian $L = D - A$. The Fiedler vector (second eigenvector) encodes a soft cut.

- Tiny $n \rightarrow$ pure shifted inverse iteration
- No external linear-algebra library
- Same TINY_GRAPH as other labs

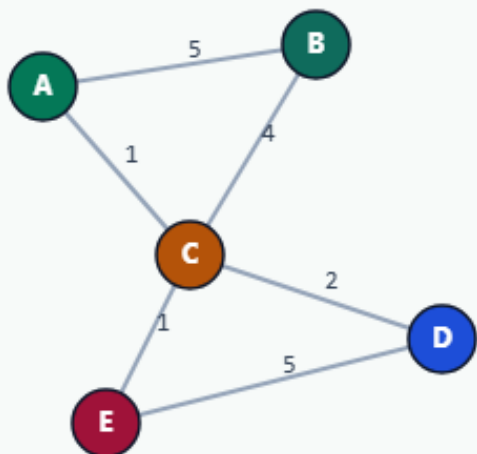
nodes: 5
edges: 6

Sort by Fiedler value

SPECTRAL BISECTION

Sort by Fiedler value

Step 2 / 5 · fiedler-order · module02-03-spectral-bisection



Explanation

After iteration, nodes order low→high as E, D, C, B, A. E is most negative; A is most positive — natural bipartition candidates.

- Order endpoints: E lowest, A highest
- Values are continuous soft memberships
- Balance filter rejects lopsided prefixes

order: E < D < C < B < A

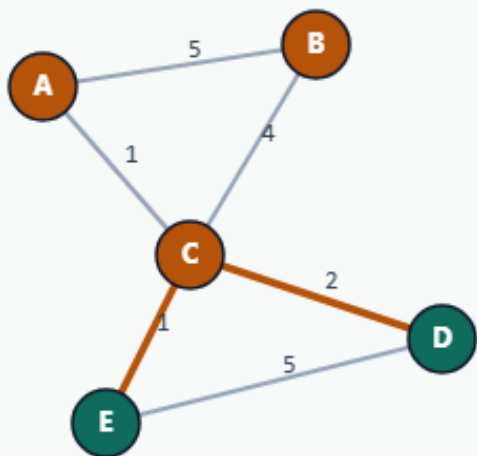
Sort by Fiedler value

Sweep prefixes for best cut

SPECTRAL BISECTION

Sweep prefixes for best cut

Step 3 / 5 · sweep-cut · module02-03-spectral-bisection



Explanation

Try every balanced prefix of the order as side 0. The winning split is {D,E} vs {A,B,C} with cutsize 3.

- Balance window $\approx 20\text{--}80\%$ size
- Pick minimum cut among legal splits
- Golden cut = 3

best parts: DE | ABC
cutsize: 3

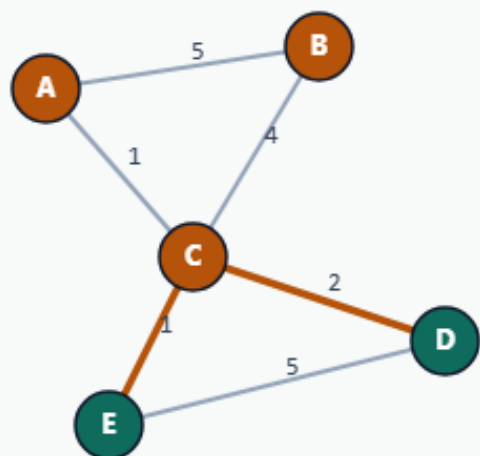
Sweep prefixes for best cut

Result ABC|DE

SPECTRAL BISECTION

Result ABC|DE

Step 4 / 5 · final · module02-03-spectral-bisection



Explanation

Spectral recovers the same communities as unconstrained greedy and LP. Teaching point: global eigenvectors can find the cut without greedy merges.

- parts: ABC|DE
- Heavy edges internal
- Bridge C-D/C-E cut

cutsizes: 3
parts: ABC|DE

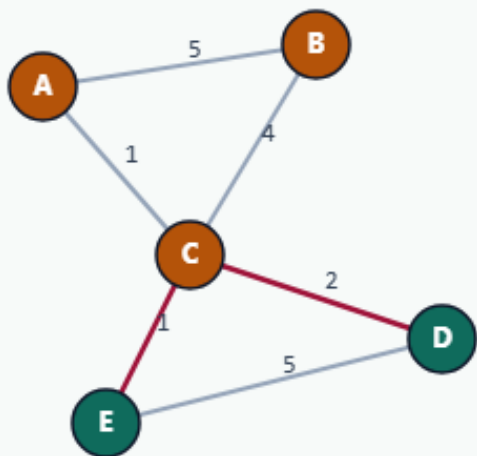
Result ABC|DE

When spectral helps

SPECTRAL BISECTION

When spectral helps

Step 5 / 5 · takeaway · module02-03-spectral-bisection



Explanation

Spectral is a strong initializer for bipartition. On larger chips it pairs with multilevel; here the tiny instance is exact enough to grade against goldens.

- Good seed for KL/FM
- Balance via sweep, not afterthought
- Next: swap/move refinement labs

Golden: cut=3, E lowest / A highest

When spectral helps

Browser lab track

- In the browser lab, run spectral bisection and inspect the Fiedler order
- Clear challenges for cutsize three and the ABC versus DE parts

Implement track

- Run the spectral mode on the tiny graph
- Confirm cutsize three and the natural clusters
- Re-implement the sweep yourself; a dense tiny eigensolve is fine at course scale

Implement track — try these

```
export PYTHONPATH=./common  
python ../common/solvers.py examples/tiny_graph.json --mode spectral
```

Pitfalls to watch

- Ignoring balance can isolate one node
- Numerical noise can scramble near-ties, use a stable sort
- Disconnected graphs need explicit component handling

Your turn

- Match cutsize three, finish the checklist and quiz, then continue to Kernighan–Lin